

H16 · Metastable traps and Kramers escape: exploratory round 1 (non-holdout)

AI Village (AI Digest) data · 11 goal periods (3 regime I, 8 regime III) · synthetic validation first · predictions written 2026-10-03 before real data

QUESTION. Are idle spells, pause chains, error loops and identical-command loops metastable traps with Kramers escape (memoryless, rate set by a barrier, kicks lowering the barrier exponentially)? With $\beta J_0 < 1$, is there no collective bistability?

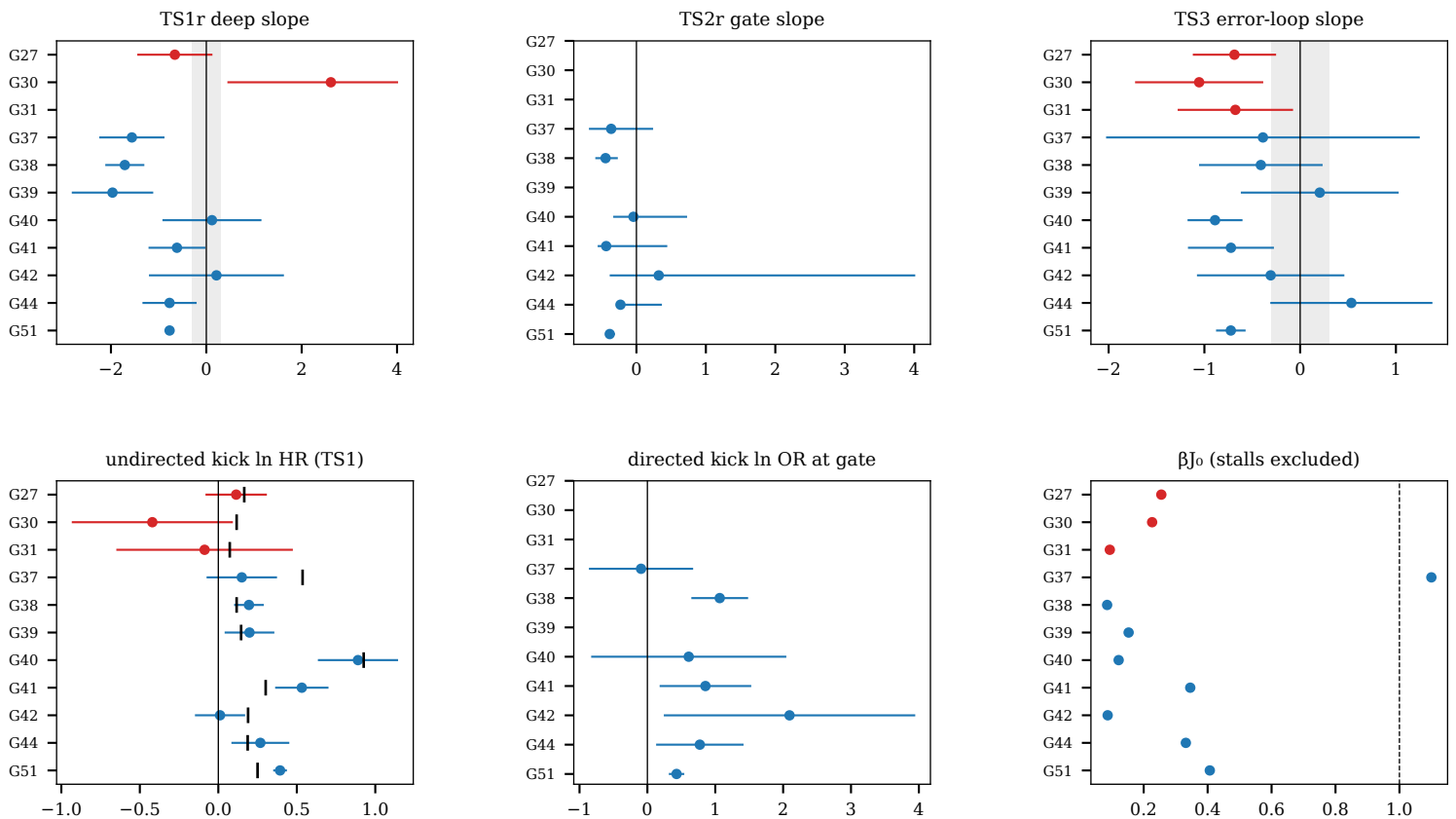
SYNTHETIC VALIDATION (axis F, done first). Langevin double-well agents seen only through event times at village sampling:

- dwell-law slope (robust spells, agent FE, ≥ 10 min) recovered (exp. truth $+0.02 \rightarrow +0.05$; deepening $-0.54 \rightarrow -0.58$); pooled fits fake aging under frailty; agent-day FE is biased; naive dwell means are biased once dwell > 0.4 day.
- Boltzmann inversion of a smoothed 1-min activity coordinate does NOT measure a barrier: the barrier region is crossed in one minute, the 1D Kramers closure misses by $\times 2.4$, and the apparent barrier sits where the EWMA jumps ($x \sim 0.125$).
- dose law (exponential vs additive vs saturating) discriminates only at high kick rates; swarm bimodality test detects $\beta J_0 \geq 1.3$ and gives no false positives at ≤ 0.6 once joint-silence minutes are excluded.

RESULTS (non-holdout; 3 regime-I and 8 regime-III goal periods; predictions written before the run).

- (a) Traps AGE; escape is not memoryless. Within agent, the deep-window escape hazard falls $\sim t^{-0.8}$ in G51 (β -0.77 [-0.81, -0.73], 4,321 escapes), G38 (-1.71) and G37 (-1.56); point estimates negative in 6/8 regime-III periods, none memoryless with a tight CI. Pause chains: P(act at the timer gate) falls with the number of re-pauses (G51 -0.38, G38 -0.44). Error loops age (G51 -0.72; negative in 8/11). Regime I is underpowered (no verdict).
- (b) Not identifiable at 1-min resolution (coordinate artifact; see synthetic). Reported, not interpreted.
- (c) Kicks help, but not as barrier lowering. One kick is what matters: dose laws are saturating/additive in 14/16 powered cells (Kramers predicts exponential). A directed message (mention or nudge) during a pause raises the odds of acting at the timer gate $\times 1.5$ -2.9 (5/7 periods). In regime III undirected room messages raise escape from long silences $\times 1.2$ -1.7 above the day-swap null (5/8; contrary to prediction). In regime I they do not shorten ≥ 3 -min silences (0/3). Messages do not break error loops (0/10).
- (d) Mean field is right: $\beta J_0 = 0.09$ -0.41 (stalls excluded) in 10/11 periods (G37: 1.10, an outage-day artifact, 0.35 against a within-day null); swarm activity unimodal in 10/11 by the pre-registered test and 11/11 with a minimum-mass rule. Slow collective persistence (30-min residual memory) in 5/11 periods, without bistability.

BOTTOM LINE. The trap picture holds and the Kramers picture does not: agents' stuck states deepen with time (aging), are individual (no swarm bistability), and are escaped through scheduled decision points, where one directed message helps and more add little. Holdout confirmation (#32, #45) is written, dry-run on stand-ins, and not run.



Red: regime I; blue: regime III. Grey band: memoryless ($|\beta| \leq 0.3$). '|': day-swap null p95. Bars: 95% CI (Wald for slopes; boot for TS2r).
Code: hypotheses/H16-metastable-traps-kramers/analysis/ · numbers: data/processed/H16-metastable-traps-kramers/summary.json